CERTIFICATE OF ANALYSIS

Refined **Castor Oil — First Special Grade (FSG)**, analysed on the manufacturer's lot certificate.

PRODUCT	Refined castor oil, first special grade (FSG)
LOT REFERENCE	14072025
MANUFACTURED	14 July 2025
RETEST / EXPIRY	14 July 2027
PACKING	200 kg HDPE drums on pallets · 16 MT net per 20-ft
PORT	FOB Mundra, India
STANDARD	As per the agreed specification for castor oil FSG

#	PARAMETER	SPECIFIED LIMIT	RESULT
1	Appearance	Pale yellow, viscous, clear liquid	**Pale yellow, viscous, clear liquid**
2	Moisture (K.F.)	0.25% max	**0.16%**
3	Free fatty acids (as oleic)	1.0% max	**0.52%**
4	Acid value (mg KOH/g)	2.00 max	**1.04**
5	Iodine value (Wijs method)	82 – 90	**84.48**
6	Saponification value (mg KOH/g)	177 – 185	**180.28**
7	Hydroxyl value (mg KOH/g)	158 – 165	**162.47**
8	Insoluble impurities	0.02	**0.00**
9	Viscosity @ 25 °C (poise)	6.0 – 9.0 cps	**7.3**
10	Refractive index @ 20 °C	1.477 – 1.481	**1.4773**
11	Specific gravity @ 30 °C	0.954 – 0.960	**0.957**
12	Identification	Positive	**Pass**
13	Odour	Characteristic, comparable to standard	**Pass**
14	Peroxide value	4.0 meq/kg	**1.67 meq/kg**
15	Ricinoleic acid	85 min	**87.45**
16	Unsaponifiable matter	1.0% max	**0.40%**
17	Colour (Gardner)	4.00 max	**3.0**
18	Colour (Lovibond, 5.25" cell)	Y 20.0 max · R 2.0 max	**Y 17.0 · R 1.7**

All parameters conform to the specification above. Results apply to the lot named on this certificate. Method references: K.F., Wijs, Gardner and Lovibond as stated.

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Mill name and signed original released to buyers on request.